

Kashyap Gorasiya

[Personal Website / Portfolio]

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GitHub: github.com/kashyap9107 [Profile]

AI Engineer

Education

MTech, Computer Science and Engineering	Indian Institute of Technology Goa	CGPA: 9.02/10	(2024–2026)
BTech, Information Technology	Dharmsinh Desai University, Nadiad, Gujarat	CGPA: 8.63/10	(2019–2023)
Class 12, GSHSEB	Bharad School, Rajkot	Aggregate: 86%	(2017–2019)
Class 10, GSHSEB	Global Science Academy, Rajkot	Aggregate: 92.33%	(2016–2017)

Experience

Deloitte – Software Engineer | Hyderabad, India (Jun 2026 – Present)

- Built a role-based issue tracker with **Angular 16**, **Spring Boot** microservices **Spring MVC**, and **22 DTO-based REST APIs** for user, project, and issue management, assignment, filtering, and status tracking. Integrated **API Gateway**, **Eureka**, **OpenFeign/RestTemplate**, **JPA/MySQL**, **BCrypt** authentication, validation, exception handling, **Swagger/OpenAPI**, and failover/load balancing.
- Deployed with **Docker** and **AWS ECR**, **ECS/Fargate**, **EC2**, and **RDS**; used **CloudWatch** for monitoring and centralized logging. Gained hands-on experience with **AI agents**. [Frontend] [Backend]

IIT Goa, CSE Department – Teaching Assistant | Goa, India (Aug 2024 – May 2026)

- Assisted in courses including **Data Structures and Algorithms**, **Algorithm Design**, **Fundamentals of Computing**, **Python**, **Operating Systems**, and **Machine Learning**.

SoftMax AI – ML Engineer (Intern) | Ahmedabad, Gujarat [Certificate] (Mar 2023 – Jun 2023)

- Fine-tuned **BERT/PyTorch** on **AWS SageMaker** for multilingual mining-risk classification across **40+ categories**. Optimized **YOLOv5/v8 (Ultralytics)**, **ByteTrack**, and **DeepSORT** for real-time vehicle detection, tracking, and counting on **EC2 GPUs**, integrating **SlowFast** into video ETL workflows.
- Built invoice-processing APIs with **AWS Textract**, **Lambda**, **Comprehend**, and **Terraform** to extract tables, line items, and key-value pairs into **structured JSON**; developed streaming ETL and real-time querying pipelines with **Glue**, **Kinesis**, **Firehose**, and **Athena**.
- Developed a **Stable Diffusion** logo-generation pipeline for **50K+ images**, combining preprocessing, quality filtering, **PyTorch fine-tuning**, and prompt engineering with **Dockerized ETL**, **S3**, and **Terraform** for reproducible ML workflows.

Ciesto Solutions – Web Developer (Intern) | Rajkot, Gujarat [Certificate] (Dec 2022 – Feb 2023)

- Developed an **e-commerce application** using **HTML**, **CSS**, **JavaScript**, **Node.js**, and **MongoDB**.

Projects

Real-Time Traffic Rule Enforcement with Drones (Jun 2025 – May 2026)

- Developed a **UAV traffic-enforcement pipeline** for rider, vehicle, helmet, and license-plate detection using semi-supervised learning. Designed **spatial association based violation detection algorithm** to match helmet violations to plates despite occlusion, scale variation, motion blur, and illumination changes.
- Enhanced **YOLOv11**, **YOLOv26**, and **RF-DETR** with **CBAM attention** and **SAHI slicing**; achieved **99.30% mAP@50** with YOLOv11m and **87.88% mAP@50:95** with RF-DETR Base on aerial imagery.
- Fine-tuned transformer-based **TESTR/UNITS** and implemented **HTS OCR** for plate recognition, using hyperparameter tuning to optimize performance. TESTR achieved **92.38% word accuracy** and **2.42% CER**; deployed the optimized **YOLO + TESTR** pipeline with approximately **104 ms end-to-end latency** for violation cases.

DocChat - Multilingual Multimodal Document RAG (Feb 2026 – May 2026)

- Engineered a **multimodal RAG pipeline** for handwritten, printed, scanned, distorted, and multilingual Indian-language documents, leveraging **Qwen3-VL 8B** for visual document understanding and semantic representation.
- Generated document embeddings, indexed them in a **FAISS vector database**, and implemented semantic similarity search with configurable **Top-K retrieval** to retrieve relevant document context for user queries.
- Integrated **LangChain** for RAG orchestration and **Llama 3.1 8B** for **context-grounded question answering**.

- Intelligent Crop Yield Prediction for Precision Farming [\[GitHub\]](#)

(Jan 2025 – Apr 2025)
- Developed a crop yield prediction and recommendation system for **22 crop types**, performing exploratory data analysis, class-wise statistical imputation, and feature engineering. Compared **KNN, Decision Tree, and SVM** using cross-validation and hyperparameter tuning; achieved **98% crop classification accuracy with SVM** (precision/recall/F1 > 0.95).

- QuickQuiz – Quiz Web Application [\[GitHub\]](#)

(Sep 2024 – Nov 2024)
- Built a **real-time multi-user quiz application** using **Django, JavaScript, HTML/CSS, Jinja, and SQLite**, implementing role-based access, cookie-based scoring, and real-time result calculation.
 - Designed **quiz creation from an existing question bank**, subject-wise quizzes, randomized questions per student, timed auto-submission, and **leaderboards, reports, and graphical analytics** across students, classes, and subjects.

- Automated Attendance System using Image Processing [\[GitHub\]](#)

(Oct 2021 – Oct 2022)
- Developed an automated classroom attendance system using **Python, TensorFlow, Flask, MTCNN**, and fine-tuned **FaceNet** on a custom facial dataset for face detection and recognition.
 - Implemented **128-dimensional face embeddings** for identity matching, achieving **83% precision**, and deployed the system on **AWS EC2** for real-time attendance processing.
 - Integrated **incremental learning** for adding new identities and automated absence notifications to parents.

Skills

Languages & Web:	Python, C++, C, Java, SQL, R, Prolog, JavaScript, HTML, CSS
Cloud & Engineering:	Microservices, Maven, Git, Docker, AWS, Terraform, MLOps, SageMaker, EC2, ECR, ECS, S3, Lambda, RDS, Textract, Comprehend, Glue, Kinesis, Firehose, Athena, ETL Pipelines
Frameworks:	Spring Boot, Spring Cloud, Spring Data JPA, Angular, PyTorch, TensorFlow, Flask, Django, LangChain, LangGraph
Databases:	PostgreSQL, MySQL, SQLite
AI/ML:	Computer Vision, NLP, Transformer Architectures, LLMs, RAG, AI Agents, Model Fine-tuning, Model Evaluation, Research Paper Reproduction
Soft Skills:	Communication, Leadership, Teamwork, Innovation, Quick Learning

Positions of Responsibility

Team Leader	Student Start-up & Innovation Policy Prototype Project with 2 members	(2021–2022)
Class Representative	Handled academic concerns of the B.Tech 2019–2023 batch	(2019–2023)
Coordinator	IIT Goa Drone Boot Camp	(May 2025)

Achievements

- Rank 1, Institute Silver Medal** in M.Tech Computer Science and Engineering, **IIT Goa** (2026)
- Competitive Programming:** GeeksforGeeks (1635 rating, 400+ problems [\[Profile\]](#)), LeetCode (1714 rating, 700+ problems [\[Profile\]](#))
- Student Start-up & Innovation Policy** Prototype completion [\[Certificate\]](#) (2023)
- Patent filed** for an AI-enabled drone system for automated traffic rule violation monitoring. [202521129482]

Extracurriculars & Hobbies

- Chess | Badminton | Books | Entrepreneurship